

ALVA, OK, USA

WMO: 722052

Lat: 36.773N

Lon: 98.670W

Elev: 1474

StdP: 13.93

Time Zone: -6.00 (NAC)

Period: 04-19

WBAN: 53933

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	10.0	15.9	0.1	5.8	16.3	5.0	7.5	21.4	32.0	47.3	28.4	49.4	10.0	0	0.604

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	24.3	102.5	72.9	100.1	73.3	98.5	73.5	78.2	93.1	77.0	91.6	75.8	90.1	15.8	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
74.6	136.6	83.3	73.0	129.6	81.8	72.4	126.7	81.1	42.8	93.4	41.5	90.3	40.3	90.0	83.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
28.8	26.1	23.7	DB	2.5	107.3	5.8	3.0	-1.7	109.4	-5.1	111.2	-8.3	112.8	-12.5	115.0	
			WB	1.5	79.6	5.5	2.0	-2.4	81.0	-5.6	82.2	-8.7	83.3	-12.7	84.8	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	59.9	36.8	39.6	49.7	57.3	68.6	79.8	83.5	80.7	74.1	60.9	47.8	38.0
	DBStd	18.86	10.15	11.56	9.89	9.31	8.76	6.06	5.79	6.05	6.91	9.46	9.63	9.68
	HDD50	1459	418	321	126	36	2	0	0	0	0	21	151	384
	HDD65	3945	873	712	478	261	65	1	1	0	9	189	519	837
	CDD50	5055	10	29	119	255	577	894	1040	952	722	360	85	12
	CDD65	2065	0	0	5	29	176	445	575	487	281	63	4	0
	CDH74	25207	3	16	76	416	2003	5662	7622	5654	2963	721	62	9
	CDH80	13485	0	1	9	123	886	3173	4563	3071	1402	245	8	4
Wind	WSAvg	11.2	10.9	11.3	12.3	12.6	10.7	12.1	10.4	10.1	11.0	11.4	11.2	10.6
Precipitation	PrecAvg	28.6	1.0	1.1	2.3	2.5	3.6	4.3	3.2	3.6	2.0	2.4	1.6	1.1
	PrecMax	36.2	3.5	3.0	7.9	8.0	10.8	8.9	6.4	7.3	5.9	9.8	5.8	3.1
	PrecMin	18.0	0.0	0.0	0.0	0.0	0.2	0.4	0.4	0.0	0.0	0.0	0.0	0.0
	PrecStd	5.3	0.9	0.9	2.0	1.8	2.2	2.3	1.5	1.9	1.5	2.5	1.8	0.9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	72.5	75.4	81.0	90.4	98.6	106.0	108.4	105.8	99.9	91.3	80.9	71.6
		MCWB	53.3	54.5	60.3	65.6	68.9	70.7	73.6	74.7	71.9	68.5	60.6	57.6
	2%	DB	63.9	70.1	75.4	82.3	93.0	100.4	104.0	101.7	95.3	86.4	74.6	63.4
		MCWB	48.8	53.5	60.2	64.1	69.6	72.6	72.5	74.3	71.7	68.4	59.0	51.5
	5%	DB	58.7	63.6	72.4	78.9	89.9	98.8	100.5	99.1	92.6	81.7	68.4	57.3
		MCWB	46.3	51.3	58.9	62.9	69.1	73.1	73.3	74.7	72.0	64.7	56.4	47.5
	10%	DB	54.2	57.2	67.8	74.6	84.2	94.9	98.8	94.7	88.5	77.2	63.7	54.1
		MCWB	44.1	47.1	56.8	61.4	67.8	72.4	73.6	73.5	71.2	63.3	53.1	46.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	57.0	59.6	67.5	70.5	76.4	79.1	79.8	79.5	79.1	75.4	65.6	60.4
		MCDB	66.9	68.0	75.3	80.5	88.8	94.5	96.8	93.6	92.2	82.7	72.9	68.0
	2%	WB	50.6	55.9	64.3	67.8	74.1	77.2	77.9	77.9	75.9	72.1	61.7	54.0
		MCDB	61.4	66.1	71.0	77.3	85.3	92.8	93.5	92.7	87.1	79.4	68.8	59.8
	5%	WB	47.1	52.5	61.5	65.8	72.4	75.8	76.6	76.5	74.1	69.1	58.7	49.1
		MCDB	58.2	63.7	67.6	75.4	82.8	90.9	91.6	91.3	85.1	78.4	65.9	56.6
	10%	WB	44.3	48.5	58.1	63.3	70.4	74.5	75.4	75.3	72.7	66.1	55.3	45.7
		MCDB	53.4	57.1	66.1	71.8	80.5	89.0	90.2	89.5	83.8	75.8	62.9	52.1
Mean Daily Temperature Range	5% DB	MDBR	23.1	23.2	24.0	24.6	23.9	25.1	24.3	22.3	23.3	24.4	24.4	21.2
		MCDBR	32.6	32.8	30.3	30.7	29.2	29.3	28.5	28.0	27.4	30.0	31.3	29.2
		MCWBR	19.9	19.7	17.6	16.9	11.6	8.6	7.7	7.9	9.5	14.8	18.4	19.0
	5% WB	MCDBR	30.8	30.2	24.0	26.4	24.1	25.6	24.0	23.0	22.4	24.6	26.5	26.8
		MCWBR	20.0	19.5	16.2	16.6	12.0	10.0	8.6	8.3	9.8	14.5	17.5	19.4
Clear-Sky Solar Irradiance	taub	0.282	0.291	0.333	0.365	0.386	0.412	0.414	0.413	0.376	0.341	0.299	0.278	
	taud	2.517	2.501	2.400	2.344	2.363	2.324	2.338	2.299	2.384	2.462	2.514	2.535	
	Ebn at Noon	292	302	296	291	285	276	274	272	278	279	283	287	
	Edh at Noon	25	29	35	39	39	41	40	40	35	30	25	23	
All-Sky Solar Radiation	RadAvg	866	1094	1443	1791	2008	2239	2206	1972	1649	1230	925	748	
	RadStd	82	148	107	107	119	181	163	120	110	103	151	108	72

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (29 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page